A Six-Layer Evaluation Stack and a Portfolio-Aware Kill Criterion for Crypto-Perp Trading Strategy Research at Laptop Scale

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May 2026

Abstract

We report a methodology — not a strategy — for evaluating systematic trading strategies on crypto perpetual futures when the practitioner is constrained to small trade counts ($N<200$), laptop-scale compute, and a single execution substrate. We argue that single-metric leaderboards (Calmar, Sharpe, or even the Deflated Sharpe Ratio of López de Prado and Bailey (2014)) over-reject genuine strategies and under-reject overfit ones at this scale, and propose a *six-layer evaluation stack* in which each layer addresses a failure mode of the one above: (L1) raw return, (L2) risk-adjusted return, (L3) sample-size awareness, (L4) multiple-testing deflation, (L5) tail and path shape, and (L6) portfolio diversification through a Marginal Diversification Benefit (MDB) triplet. The load-bearing contributions are: a *portfolio-aware* hard-MDD kill criterion (K1) that corrects the over-rejection induced by single-strategy pre-registration when adding a candidate to an existing book, anchored on combined-book MDD, robust MDB, and maximum pairwise Pearson; a demonstration that *continuous position shrinkage* along a bull-return / bear-drawdown Pareto frontier dominates binary regime gates, exhibited via a V1/V2/V3 sweep of an HMM-and-slope strategy; and a candid treatment of DSR over-rejection at $N<200$ with kurtosis 5–340 typical of trend strategies with fat right tails. We apply the framework to a candidate book of two strategies (T3 `SmaRegime180`, R$\wedge$T2 `HmmSmaSlopeV2`) with pairwise Pearson correlation 0.07, combined-book risk-parity MDD 2.12%, and robust MDB +0.55. We name the ablations: pairs/cointegration (X1) and funding mean-reversion (F1) fall out of the framework cleanly. The paper is paired with a forward held-out window (Binance 2026-06 → 2026-12) that has not been downloaded and will not be until after submission; results on it will be reported verbatim.

1 Introduction

Backtest-driven strategy research at laptop scale is a constrained optimisation problem: thin trade counts, a single execution substrate (here, Binance USDT-margined perpetual futures via Freqtrade (Freqtrade contributors, 2026)), a multi-year but still finite history, and the practitioner's own search process generating selection bias. The pre-registration discipline of Bailey et al. (2014) mitigates the last of these; the Deflated Sharpe Ratio of López de Prado and Bailey (2014) addresses multiple-testing inflation. Both are necessary; neither, applied alone, is sufficient. In our setting (5.5 years of Binance perpetuals, 5 coins, N closed trades typically in $[30, 700]$, returns with sample skew up to +21 and excess kurtosis up to +470), DSR flags every candidate strategy as "noise." If we gate on DSR alone, the project is over. If we ignore it, we replay the same overfitting failures that motivated its introduction.

This paper documents what we did instead. We construct a six-layer stack (§3) in which each layer addresses a failure mode of the one above. Critically, we treat L4 (DSR) as a *humility check* rather than a binding gate, and add L5 (tail and path shape) and L6 (portfolio diversification) as load-bearing gates. We then formalise a *portfolio-aware* hard-MDD kill criterion (§4) — the single most important methodological correction in the paper — that prevents a pre-registered single-strategy kill threshold from over-rejecting a candidate whose drawdown is uncorrelated with the existing book.

1.1 Contributions

1. **A six-layer evaluation stack** (L1–L6) in which each layer addresses a specific failure mode of the previous layer. The stack is portable: the layer count is fixed, but the gate thresholds at L2, L3, L5, and L6 are calibrated to the substrate.

2. **A portfolio-aware K1 kill rule** (§4) with four clauses — combined-book MDD, robust MDB triplet (equal-weight, risk-parity, mean-var), max pairwise Pearson, and a $2 \times K1$ hard cap. We show that the standalone K1 of Algo-traders project (2026a) over-rejects when applied as a portfolio admission rule, and the corrected gate admits the second strategy in our candidate book on principled grounds.

3. **Continuous shrinkage along a Pareto frontier** as a dominant alternative to binary regime gates. A V1/V2/V3 sweep of the HMM-and-slope strategy (§6) traces a continuous bull-return / bear-drawdown frontier rather than collapsing to a single operating point.

4. **Empirical scaffolding** for a candidate two-strategy book {T3 `SmaRegime180`, R∧T2 `HmmSmaSlopeV2`} on Binance 5-coin perpetuals 2020-09 → 2026-05, with pairwise Pearson 0.07 and robust MDB-rp +0.55. Five strategy families that *fail* the stack (X1 pairs, F1 funding MR, C1 funding carry, R2 HmmRegime4Rolling, R∧C1 HmmCarry conjunction) function as ablations of the framework.

1.2 What this paper is not (and why)

This paper is not a strategy proposal. The candidate book is the *worked example* of the methodology, not its endpoint. It is also not yet an out-of-sample claim: the forward window 2026-06 → 2026-12 has intentionally not been downloaded, and the 30-day live paper-trade dry-run is unfinished. We are explicit about both gaps (§7). Finally, this is not a critique of López de Prado and Bailey (2014): DSR is the correct gate at institutional N; it is the wrong gate at laptop N with heavy right-tailed distributions. The contribution is a framework that names the regime in which DSR is informative versus binding.

Reproducibility. All backtests use Freqtrade with `--fee 0.00035` (Hyperliquid taker, applied uniformly across the universe for cross-strategy comparability), public Binance perpetuals OHLCV and funding data via `ccxt`, and the common evaluation substrate documented in Algo-traders project (2026b). Source code, decision documents, and per-strategy result cards are in the companion repository `algo-traders`.

2 Related Work

2.1 Backtest evaluation and overfitting

The line of work descending from Bailey et al. (2014) and López de Prado and Bailey (2014) formalises two distinct failure modes of naive backtest evaluation. The Probability of Backtest

Overfitting (PBO) captures the selection-bias problem: an analyst running K trials and reporting the best Sharpe systematically overstates out-of-sample performance. The Deflated Sharpe Ratio (DSR) corrects the observed Sharpe for: (a) the number of trials, (b) the cross-trial Sharpe variance, and (c) the skew and kurtosis of the strategy returns. The skew/kurtosis correction is the term that bites at our scale — with kurtosis 5–340 our raw Sharpe estimates are aggressively shrunken even before the multiple-testing adjustment, and the resulting DSR is below conventional significance for every standalone strategy we tested. We retain DSR as Layer 4 (§3.4) but do not gate admission on it.

Simhi et al. (2026) is the related warning from the LLM-assisted strategy generation regime: generators trained on retail trading content recycle discredited indicators (e.g. RSI thresholds, MACD crossovers) because they dominate the training corpus. The lesson translates: a layered gate with explicit second- and third-moment terms is robust to the "looks-like-a-strategy" failure mode in a way that a single-metric gate is not.

2.2 Crypto-perp microstructure and funding

Le et al. (2026) provides the funding-rate Ornstein-Uhlenbeck calibration we adopt as a default prior for funding-driven strategies, with empirical half-life $\approx$ 8h on Hyperliquid majors. Badawi et al. (2025) and Inan (2025) establish short-horizon predictability of funding rates that justifies a 4h cadence funding-aware strategy; Anonymous (2026c) and Anonymous (2025a) document the CEX/DEX two-tier structure that determines whether funding is a return source or a cost. Anonymous (2026b) provides Hyperliquid-specific slippage distributions that inform our fee assumption (`--fee 0.00035` is the Hyperliquid taker; applying it uniformly to Binance backtests is a conservative bias). The funding mean-reversion strategy F1 (§5.4) is the failed test of the funding-extreme thesis on our substrate; it falls out of the framework cleanly.

2.3 Regime detection

Pang et al. (2026) applies Gaussian HMMs to Bitcoin daily returns; Anonymous (2026d) extends to a Wasserstein-distance-based variant that is more robust to non-Gaussian state emissions. Our R-family strategies (`HmmRegime4`, `HmmRegime4Rolling`) implement a standard 4-state Gaussian HMM on 1h returns. Anonymous (2026a) documents the "regime-timing" gap between backtest and live performance, which motivates treating $\sim$ (look-ahead) HMM variants as unbeatable ceilings, not strategies. Anonymous (2025b) establishes the 1h–4h timescale as the boundary at which trend and mean-reversion structures co-exist; this is the scale at which we run.

3 The Six-Layer Evaluation Stack

We define six layers, each posed as a question and answered by an explicit metric. Each layer addresses a specific failure mode of the previous one (Table 1).

The reading rule we follow: a strategy is admitted to paper-trade if it is positive on L2–L3–L5 *and* has robust positive MDB-rp at meaningful magnitude against the current book. L4 (DSR) is reported but not gating; §3.4 explains why.

3.1 Layer 1: Return

L1 is total return (or CAGR). It is the layer below which nothing matters and above which nothing is decided. We report it but never gate on it.

Layer	Question	Headline metric	Fixes failure of
L1	Profitable?	CAGR	—
L2	Risk-adjusted?	Calmar, Sharpe	L1: scale-free
L3	Sample-size aware?	SQN	L2: thin samples
L4	Multiple-testing?	DSR	L3: trial inflation
L5	Tail/path?	Ulcer, Martin, kurtosis, $\mathrm{CVaR}_{5\%}$	L2: Gaussian lies
L6	Portfolio-additive?	MDB-rp, corr	L2–L5: redundancy

Table 1: The six-layer evaluation stack. Each layer fixes a failure mode of the previous layer; the stack as a whole gates admission to the candidate book.

3.2 Layer 2: Risk-adjusted return

Calmar (CAGR over maximum drawdown) and **Sharpe** (annualised return over annualised volatility, 365-day basis). Calmar is our headline because drawdown rather than volatility is the binding constraint at laptop scale — a 20% drawdown ends the experiment regardless of long-run expectation. Sharpe is reported for cross-paper comparability.

3.3 Layer 3: Sample-size awareness

The System Quality Number,

$$\mathrm{SQN} \;=\; \frac{\bar{r}}{\sigma_r} \cdot \sqrt{N}, \tag{1}$$

where $\bar{r}$ and σ_r are the mean and standard deviation of *per-trade* returns and N is the trade count. SQN penalises strategies whose Calmar is driven by a small number of trades. In our universe, $\mathrm{SQN} \geq 1.7$ on $N \geq 80$ trades is the empirical threshold below which standalone Calmar should not be trusted as a point estimate.

3.4 Layer 4: Multiple-testing deflation — and when it over-rejects

The Deflated Sharpe Ratio of López de Prado and Bailey (2014) is

$$\mathrm{DSR} \;=\; \Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}_0)\sqrt{N-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{\mathrm{SR}}^2}}\right), \tag{2}$$

where $\widehat{\mathrm{SR}}$ is the observed Sharpe, SR_0 is a benchmark adjusted for the number of trials and cross-trial Sharpe variance, $\hat{\gamma}_3$ and $\hat{\gamma}_4$ are sample skew and kurtosis of returns, and Φ is the standard normal CDF.

Two terms in the denominator dominate at our scale. First, $\hat{\gamma}_4$ ranges from 5 to 340 across our strategies; the kurtosis correction shrinks the test statistic by a factor of 3–20. Second, $\sqrt{N-1}$ with $N \in [30, 700]$ does not overcome the variance inflation. The empirical result, documented in our DSR sweep (Algo-traders project, 2026c), is that *every* standalone strategy receives DSR below conventional significance, including strategies with Calmar > 8 on > 90 trades across two bull/bear cycles. Gating on DSR alone retires the entire project.

We adopt the following rule: DSR is computed and reported on every result card, but admission is gated on L2, L3, L5, and L6 jointly. DSR functions as a humility check — a signal to look harder at L5 (tail shape) and L6 (portfolio additivity), not a binary kill. This is the empirical analogue of the survival-of-the-quant comment in Davies (2025): a practitioner who only ships strategies with $\mathrm{DSR} > 0.5$ at $N = 100$ with kurtosis 50 ships nothing.

3.5 Layer 5: Tail and path shape

L2 reports the worst single drawdown; L5 reports the *shape* of the underwater experience.

The **Ulcer Index** of the equity curve $\{W_t\}$ is

$$\text{Ulcer} \;=\; \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(\frac{W_t - \max_{s\le t} W_s}{\max_{s\le t} W_s}\right)^2 \cdot 100^2}. \tag{3}$$

Ulcer is path-aware: a strategy with a single deep drawdown that recovers quickly has low Ulcer; a strategy with a chronic shallow drawdown has high Ulcer despite a smaller MDD. The Martin ratio is Martin = CAGR/Ulcer.

We additionally report sample **skew** ($\hat\gamma_3$), excess **kurtosis** ($\hat\gamma_4$), tail ratio $|P_{95}|/|P_5|$, and $\text{CVaR}_{5\%} = \mathbb{E}[r \mid r \le P_5]$ of daily log-returns. Slope-gated strategies in our universe have $\hat\gamma_3 \in [+9, +21]$ and $\hat\gamma_4 \in [+110, +470]$. Sharpe overstates these by a Jensen-inequality margin that L5 makes visible.

Worked comparison. T3 (`SmaRegime180`) has MDD 2.21%, Ulcer 1.30, Pain 1.12 — a tight, fast-recovering path. R2 (`HmmRegime4Rolling`) has MDD 7.65% with Ulcer 4.01, Pain 3.47 — chronic underwater. The MDD gap (2.21% vs 7.65%) understates the path-shape gap; L5 makes the gap binding.

3.6 Layer 6: Portfolio diversification

L1–L5 evaluate strategies in isolation. L6 evaluates them as additions to a current book.

Let $B = \{s_1, \ldots, s_k\}$ be the current book and c a candidate. Define the equity curve of book X under weighting scheme w as $E_w(X)$ and the Sharpe as $\text{SR}(\cdot)$. The **Marginal Diversification Benefit** is

$$\text{MDB}_w(c, B) \;=\; \text{SR}(E_w(B \cup \{c\})) \;-\; \text{SR}(E_w(B)). \tag{4}$$

We compute MDB under three weighting schemes:

- **MDB-eq:** equal-weight, $w_i = 1/|B \cup \{c\}|$.
- **MDB-rp:** risk-parity. Trailing 90-day return volatility σ_i per strategy; weights $w_i \propto 1/\sigma_i$, renormalised. This is the leaderboard headline.
- **MDB-mv:** mean-variance, long-only quadratic optimisation. Reported as an upper bound; known to be unstable at small N.

A candidate is **robustly** diversifying iff MDB > 0 under all three schemes. The robustness flag is the load-bearing protection against weighting-artifact false positives: a strategy whose only positive MDB comes from the mean-variance scheme is being rescued by optimiser noise, not by genuine diversification.

Headline result. The candidate book {T3, R∧T2} has pairwise Pearson correlation 0.07; R∧T2's MDB-rp against {T3} alone is +0.55, robust positive under all three schemes.

4 Portfolio-Aware Kill Criteria

4.1 The single-strategy hard kill (K1)

Decision document 004 (Algo-traders project, 2026a) pre-registers a hard-MDD kill threshold for the T3 `SmaRegime180` strategy:

$$K1_{\text{standalone}} \;=\; 5.5\% \;\approx\; 3 \cdot \text{IS-MDD}(T3, \text{Hyperliquid bear}). \tag{5}$$

The 3× multiplier captures the empirical observation that live maximum drawdown typically exceeds in-sample MDD by a factor of three under regime shift. This is the standalone gate: if any single strategy in the book breaches 5.5% live MDD, it is retired.

4.2 Why standalone K1 over-rejects portfolio additions

R∧T2 `HmmSmaSlopeV2` has standalone MDD 6.05% on the common window — 0.55 percentage points above K1. Under a strict reading, R∧T2 is killed and the book collapses to a single strategy. Two implicit assumptions in the standalone K1 derivation do not transfer:

1. **Single-window scope.** The 1.74% IS anchor was measured on one 6-month BTC-only bear. On the 5.5-year 5-coin Binance common window T3's *own* MDD widens to 2.21% — the same strategy on a more demanding substrate. Consistent application of the 3× rule places K1 at $\approx$ 6.6% in the common-window basis.

2. **Single-strategy scope.** A 6.05% drawdown that occurs at a *different time* from T3's drawdowns is a fundamentally different risk object from a 6.05% drawdown that compounds T3's. At pairwise Pearson 0.07, the combined-book MDD under risk-parity weighting is bounded well below the sum.

4.3 The portfolio-aware gate

A candidate strategy c may enter a book B (where B already contains at least one standalone-K1-compliant "anchor" strategy) if *either* the standalone path or the portfolio path is satisfied:

$$\textbf{Standalone path:}\quad \text{MDD}(c) \le K1_{\text{standalone}} \tag{6}$$

$$\begin{aligned}\textbf{Portfolio path:}\quad & \text{MDD}(c) \le K1_{\text{hard cap}} = 2 \cdot K1_{\text{standalone}} \\ \wedge\; & \text{MDD}(B \cup \{c\}) \le K1_{\text{book}} = K1_{\text{standalone}} \\ \wedge\; & \text{MDB}_w(c, B) > 0 \;\; \forall w \in \{\text{eq}, \text{rp}, \text{mv}\} \\ \wedge\; & \text{MDB}_{\text{rp}}(c, B) \ge 0.30 \\ \wedge\; & \max_{s \in B} |\rho(c, s)| < 0.85.\end{aligned} \tag{7}$$

Each clause has a specific failure mode it addresses. The hard cap $2 \cdot K1$ prevents arbitrarily large standalone drawdowns from being "rescued" by trivially low risk-parity weight (§4.5). The book-MDD clause ensures the portfolio's own drawdown still respects the original ceiling. The robust MDB triplet prevents weighting-artifact admission. The MDB-rp magnitude threshold $\geq$ 0.30 requires a meaningful diversification benefit, not noise-level positivity. The Pearson clause prevents statistical duplicates (e.g. R∧T1/V2/V3 collectively, with mutual correlations 0.96–1.00, cannot all be in the book).

Clause	Required	Observed	Pass
$\mathrm{MDD}(c) \leq 5.5\%$	5.5%	6.05%	×
→ falls through to portfolio path			
$\mathrm{MDD}(c) \leq 11.0\%$	11.0%	6.05%	✓
$\mathrm{MDD}_{\mathrm{rp}}(B \cup \{c\}) \leq 5.5\%$	5.5%	2.12%	✓
robust MDB > 0 (eq, rp, mv)	all > 0	all > 0	✓
$\mathrm{MDB}_{\mathrm{rp}} \geq 0.30$	0.30	+0.55	✓
$\max \rho(c, B) < 0.85$	0.85	0.07	✓

Table 2: Portfolio-aware gate, applied to R∧T2 entering book {T3}.

4.4 R∧T2 walkthrough

Applied to candidate R∧T2 against book $B = \{T3\}$:

The combined-book MDD of 2.12% is computed under risk-parity weights at window endpoint $\{T3 : 0.685, R \wedge T2 : 0.315\}$ on the 2054-day common window. The two strategies' drawdowns occur at different times, and the risk-parity weighting caps R∧T2's contribution; the combined book sits at less than 40% of the original gate.

4.5 Bounds on the relaxation

The portfolio-aware gate is not a license for arbitrarily larger standalone drawdowns. Three boundary conditions are explicit:

1. **Hard cap $K1_{\text{hard cap}} = 2 \cdot K1_{\text{standalone}}$.** Above this, no portfolio justification rescues the candidate. At MDD $> 11\%$, the only way risk-parity can neutralise the strategy's contribution is to drive its weight toward zero — a "phantom entry" that inflates the leaderboard without changing live P&L. R2x (`HmmRegime4Rolling` 5-coin, MDD 21.47%) is the worked example: technically MDB-positive but excluded for precisely this reason.

2. **No daisy-chaining.** A strategy admitted via the portfolio path does not itself become an anchor. The book must always contain at least one standalone-K1-compliant strategy.

3. **Cross-cycle requirement.** Portfolio-path admission requires the common window to span at least one bull and one bear regime (here, two of each). A K1 breach measured on a single regime cannot be rescued by an MDB that is itself regime-conditional.

5 Empirical Application

5.1 Data

Binance USDT-margined perpetual futures, 5 coins (BTC, ETH, SOL, AVAX, DOGE), 4h and 1h bars, common window 2020-09-23 → 2026-05-09 (2054 days, 2 bulls and 2 bears). Fees `--fee 0.00035` (Hyperliquid taker, applied uniformly). Funding rates downloaded for the full 5.5-year window in May 2026 (superseding an earlier 2.3-year truncated parquet). Data sourcing via `ccxt`; pipeline in the companion repository.

Code	Strategy	Calmar	SQN	MDD%	Ulcer	Martin	corr→T3	MDB-rp	Status
T3	SmaRegime180	8.76	1.78	2.21	1.30	+2.51	(book)	(book)	⋆
R∧T2	HmmSmaSlopeV2	30.23	2.73	6.05	2.87	+7.41	(book)	(book)	⋆
R∧T3	HmmSmaSlopeV3	27.28	2.79	6.91	3.30	+6.58	+0.07	−0.023	▲
R∧T1	HmmSmaSlope (V1)	25.01	2.97	8.21	3.82	+6.00	+0.07	+0.012	▲
R1∼	HmmRegime4 (look-ahead)	9.16	3.88	2.94	1.09	+4.24	0.00	+0.07	∼
T2	SmaRegime720	5.39	1.74	3.57	1.75	+1.82	+0.65	−0.023	▲
T1	TrendFilter200	1.69	1.10	8.54	3.79	+0.67	0.00	+0.040	▲
X2	CrossSectionalMom.	—	—	13.04	—	—	0.00	+0.048	▲
R2x	HmmRegime4Rolling 5c	3.79	1.66	21.47	10.25	+1.15	−0.01	+0.069	×
R2	HmmRegime4Rolling BTC	0.47	0.39	7.65	4.01	+0.17	+0.01	+0.010	×
X1	PairsZScore (1-leg)	2.08	0.95	4.03	0.48	+0.55	0.00	−0.898	×
X1v2	PairsZScoreV2 (2-leg)	0.97	0.42	2.91	0.82	+0.65	n/c	n/c	×
C1	FundingCarry 5c (full)	1.93	1.22	15.65	5.87	+0.81	n/c	n/c	×
R∧C1	HmmCarry 5c	0.08	0.14	35.46	9.57	+0.04	0.00	−0.000	×
R∧C1^r	HmmCarryReverse 5c	0.78	0.56	15.09	7.50	+0.28	n/c	n/c	×
F1	FundingExtremeMR 5c	−0.32	−0.64	33.51	15.98	−0.14	n/c	n/c	×
T0	LongOnlyStrategy	−0.37	−0.58	10.17	6.26	−0.11	0.00	−0.009	·

Table 3: Common-window leaderboard. Binance perp, 5-coin universe (BTC/ETH/SOL/AVAX/-DOGE), 4h or 1h bars, fees `--fee 0.00035` uniformly. Status: ⋆ admitted to book; ▲ research frontier; ∼ look-ahead upper bound; × killed; · baseline. "n/c" = not computed (book composition unchanged).

5.2 Strategy universe

Seven families, distinguished by signal source and conjunction structure: **T** (trend, SMA + slope), **R** (regime, HMM posterior), **C** (carry, funding rate), **R∧T** (regime × trend conjunction), **R∧C** (regime × carry), **X** (cross-sectional / pairs), **F** (funding mean-reversion). Each strategy carries a pre-registered kill-criteria document (§4.5, (Algo-traders project, 2026a,b)).

5.3 The candidate book {T3, R∧T2}: per-layer evidence

Table 4 reports the per-layer evidence for the two admitted strategies.

5.4 Ablations: families that fall out of the framework

Five strategy families fail the stack at distinct layers. Each functions as an ablation: the layer at which they fail identifies the failure mode the layer was constructed to catch.

X1 (pairs/cointegration). Single-leg `PairsZScore` SOL-DOGE has standalone Calmar 2.08 and MDD 4.03%. It passes L1–L5 and the standalone K1. It fails L6 catastrophically: $\mathrm{MDB}_{\mathrm{rp}} = -0.898$. The low-vol return profile receives oversized risk-parity weight without enough return to justify it; the book's Sharpe drops when X1 is added. The two-leg variant X1v2 (Algo-traders project, 2026g) reaches the same kill verdict through a different failure: Freqtrade cannot atomically pair-stop on spread (per-leg stops fire on per-instrument prices), so the legs are not co-managed and the strategy is a directional bet on each leg separately. This is a framework limit, not a strategy failure (§7.4).

Layer / Metric	T3 `SmaRegime180`	R∧T2 `HmmSmaSlopeV2`
Universe	BTC 4h	5-coin 4h
L1 CAGR	+3.42%	+21.31%
L2 Calmar	8.76	30.23
L2 MDD	2.21%	6.05%
L3 SQN	1.78	2.73
L3 trades	85	616
L5 Ulcer	1.30	2.87
L5 Martin	+2.51	+7.41
L5 skew	+18	+11
L5 kurtosis (excess)	+440	+115
L6 $\rho \to$ other	0.07	0.07
L6 MDB-rp vs other	+0.55 (book)	+0.55 (book)

Table 4: Per-layer evidence for the candidate book. T3 is the conservative anchor; R∧T2 is the high-return / low-correlation complement admitted via the portfolio-aware gate.

F1 (funding mean-reversion). `FundingExtremeMR` fails L1 and L2: −11.60% return, Calmar −0.32, MDD 33.51% on the full 5.5-year window (Algo-traders project, 2026e). Counter-funding does not mean-revert at 4h cadence on Binance majors; the F1 thesis fails on this substrate.

C1 (funding carry). `FundingCarry` 5-coin shows genuine standalone edge after the full-window funding parquet: +32.44% return, Calmar 1.93, MDD 15.65% (Algo-traders project, 2026d). It fails the portfolio path on the hard cap clause: MDD 15.65% $> 11\% = 2K1$. The L6 evaluation is moot — C1 cannot be in the book regardless of its MDB.

R2 (HmmRegime4Rolling). Walk-forward HMM on BTC 1h. Calmar 0.47, MDD 7.65%, Ulcer 4.01 — the L5 worked comparison from §3.5. Fails L2 (Calmar) and the standalone K1. Look-ahead variant R1 has Calmar 9.16 on the same substrate; the 0.47-to-9.16 ratio quantifies the look-ahead absorption. R2 is a clean ablation of the "honest walk-forward versus look-ahead" gap that motivates pre-registration.

R∧C1 (HmmCarry conjunction). The product of an HMM-bull signal and a funding-negative signal. Hypothesis: independent signals should intersect to a tighter filter. Empirically the signals are *anti-complementary* — HMM is reactive (lags regime change) while funding is forward-looking (leads). The intersection picks the worst of both worlds: late-cycle bull lag with early-cycle bear lead. Calmar 0.08, MDD 35.46%. The reverse-sign variant R∧C1^r (HMM-bull AND funding-positive) is directionally correct but still breaches K1 by 2.7× (Algo-traders project, 2026f). Both confirm that signal-conjunction arguments require empirical verification of independence, not assertion.

6 Continuous Shrinkage along the Pareto Frontier

6.1 The V1/V2/V3 sweep

We constructed three variants of the HMM-and-slope strategy (R∧T1, R∧T2, R∧T3) differing only in their position-sizing function applied to the HMM bull-state posterior p_t:

$$
\begin{aligned}
\textbf{V1 (binary):} \quad & h_t = \mathbb{1}[p_t > \tau], \\
\textbf{V2 (linear):} \quad & h_t = \max(0,\, p_t - \tau), \\
\textbf{V3 (sqrt):} \quad & h_t = \max(0,\, \sqrt{p_t - \tau}).
\end{aligned}
$$

Each variant traces a different point on the bull-return / bear-drawdown Pareto frontier. The full result triplet (Table 5) shows the frontier directly.

Variant	Calmar	SQN	MDD%	Martin
V1 (binary)	25.01	2.97	8.21	+6.00
V2 (linear)	30.23	2.73	6.05	+7.41
V3 (sqrt)	27.28	2.79	6.91	+6.58

Table 5: V1/V2/V3 sweep. Pairwise correlations 0.96–1.00 (statistically one strategy); V2 selected as portfolio representative on best Calmar/MDD pair.

The three variants have pairwise Pearson correlations 0.96–1.00: they are *statistically one strategy* measured at three points on a shrinkage curve. The L6 framework correctly excludes V1 and V3 from the book once V2 is admitted (Pearson clause; Table 3). They remain on the research frontier as boundary points.

6.2 Reading the frontier as a continuous knob

The binary-versus-continuous-sizing question generalises beyond the V-sweep. Following Ravagnani et al. (2026), we treat

$$h_{\text{robust}} = h_{\text{standard}} \cdot \left(1 - \frac{\delta}{\sigma^2}\right) \tag{8}$$

as the canonical continuous-shrinkage form, where δ is the magnitude of the regime-correction estimate and σ^2 is its sampling variance. Binary gates ($h \in \{0, 1\}$) are the limit of this rule as $\sigma^2 \to 0$; they are statistically wasteful when σ^2 is finite and the regime signal is noisy. Decision 004 already encodes this in the `calmar_factor`, `pf_factor`, `regime_factor` multipliers (Algo-traders project, 2026a); the V-sweep is the empirical confirmation that the continuous form sits strictly above the binary form on the Pareto frontier on this substrate.

7 Threats to Validity

7.1 Held-out window not yet evaluated

The forward window Binance 2026-06-01 → 2026-12-31 has intentionally not been downloaded as of the submission date. This is the single most important unrun gate. The pre-commitment is explicit: download once after submission, run the book and the surviving frontier strategies without parameter tweaks, report results verbatim. No claim in this paper is robust until this gate is crossed.

7.2 Laptop-scale N — DSR versus evidence tension

We argue (§3.4) that DSR over-rejects at $N < 200$ with kurtosis > 50. This is an empirical claim about our substrate, not a theoretical critique of the DSR derivation. A reviewer might reasonably reverse the conclusion: *exactly* these strategies are the ones the DSR construction is designed to catch, and the framework's L5/L6 admissions are simply laundered overfitting. We do not have a clean refutation; the held-out window is the strongest available test, and we have not run it. We name the tension rather than resolve it.

7.3 Single-substrate framework calibration

All gate thresholds (K1 = 5.5%, hard cap = 11%, MDB-rp magnitude floor 0.30, Pearson ceiling 0.85) are calibrated on Binance 5-coin perpetuals 2020-09 → 2026-05. The layer structure is portable; the specific scalars are substrate-conditional. Application to a different exchange, asset class, or time horizon requires recalibration via the same pre-registration discipline that produced the original values.

7.4 Framework limit: Freqtrade cannot atomically pair-stop on spread

The X1v2 two-leg PairsZScore result (Algo-traders project, 2026g) surfaces a specific limit of the substrate. Freqtrade's per-leg stop-loss machinery fires on each instrument's own price, not on the joint spread; a divergence that widens one side's price closes that leg independently of the joint z-score, breaking market-neutrality. This is a framework limit, not a strategy failure: a true pairs evaluation requires a different execution substrate (Backtrader, Lean, or custom paired risk management above Freqtrade). The X1 family verdict in this paper is therefore narrower than "pairs do not work on crypto majors": it is "pairs cannot be faithfully evaluated within this substrate."

7.5 LLM-recycled strategy templates

Following Simhi et al. (2026): this paper's strategies were hand-coded, but the family taxonomy and metric choices were informed by LLM-assisted literature triage. The risk that load-bearing methodological choices reflect training-corpus consensus rather than principled derivation is non-zero. The L4 humility check and the explicit pre-registration template are the available mitigations.

8 Future Work

8.1 Forward held-out window 2026-06 → 2026-12 (pre-registered, frozen)

Run the candidate book and the surviving research-frontier strategies on 2026-06-01 → 2026-12-31, one-shot, no parameter tweaks. Report verbatim. This is the single highest-priority follow-up.

8.2 30-day live paper-trade dry-run

Run $\{$T3, R$\wedge$T2$\}$ on Hyperliquid in paper-trade mode for 30 days against the decision-004 kill criteria before any real capital. This catches the backtest-to-live divergence documented in Anonymous (2026a) that the held-out window cannot.

8.3 Per-coin signed-funding extension

The R$\wedge$C family (HMM conjunction with funding) is killed in both sign directions on a single-rule basis. AVAX shows opposite sign to BTC/ETH/SOL/DOGE in our data, suggesting per-coin signed-funding learning is the live extension. We leave this to future work.

9 Conclusion

We have presented a six-layer evaluation stack for crypto-perp trading strategy research at laptop scale, a portfolio-aware kill criterion that corrects single-strategy over-rejection, an empirical confirmation that continuous shrinkage dominates binary gates along the relevant Pareto frontier, and a candid treatment of where DSR over-rejects at our N and kurtosis. The candidate book $\{$T3, R$\wedge$T2$\}$ with pairwise Pearson 0.07 and combined-book risk-parity MDD 2.12% is the worked example; the five ablations (X1, F1, C1, R2, R$\wedge$C1) name the layers at which strategy families fall out.

The paper is incomplete by design: the forward held-out window and the live paper-trade dry-run are the two binding tests we have not run. We commit to running both before any claim in this paper is treated as out-of-sample robust. The framework is the contribution; the candidate book is its current output, not its endpoint.

References

Algo-traders project. Decision 004: Pre-registered kill criteria for SmaRegime180, 2026a. `backtesting/wiki/decisions/004-kill-criteria-sma-regime-180.md`.

Algo-traders project. Decision 005: Evaluation tooling and diversity plan, 2026b. `backtesting/wiki/decisions/005-evaluation-and-diversity-plan.md`.

Algo-traders project. Dsr analysis sweep, 2026c. `backtesting/wiki/results/2026-05-10-dsr-analysis.md`.

Algo-traders project. Result card: FundingCarry 5-coin full window, 2026d. `backtesting/wiki/results/2026-05-16-funding-carry-fullwin.md`.

Algo-traders project. Result card: FundingExtremeMR 5-coin full window, 2026e. `backtesting/wiki/results/2026-05-16-funding-extreme-mr-fullwin.md`.

Algo-traders project. Result card: HmmCarryReverse, 2026f. `backtesting/wiki/results/2026-05-16-hmm-carry-reverse.md`.

Algo-traders project. Result card: PairsZScoreV2 two-leg, 2026g. `backtesting/wiki/results/2026-05-16-pairs-zscore-v2.md`.

Anonymous. Exploring risk and return profiles of funding rate arbitrage on CEX and DEX. *Blockchain: Research and Applications (Elsevier)*, 2025a.

Anonymous. Trends and reversion in financial markets on time scales from minutes to decades. *arXiv preprint arXiv:2501.16772*, 2025b.

Anonymous. Evaluating structured strategy backtests: Peer benchmarks, regime timing, and live performance. *arXiv preprint arXiv:2604.18821*, 2026a.

Anonymous. Slippage-at-risk (SaR): A forward-looking liquidity risk framework for perpetual futures exchanges. *arXiv preprint arXiv:2603.09164*, 2026b.

Anonymous. The two-tiered structure of cryptocurrency funding rate markets. *MDPI Mathematics*, 14(2):346, 2026c.

Anonymous. Explainable regime-aware investing via wasserstein HMM. *arXiv preprint arXiv:2603.04441*, 2026d.

M. Badawi et al. Who sets the range? funding mechanics and 4h context in crypto markets. *arXiv preprint arXiv:2601.06084*, 2025.

David H. Bailey, Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu. The probability of backtest overfitting. *Journal of Computational Finance*, 2014. Also SSRN 2326253.

Dan Davies. On the survival-of-the-quant problem (informal note), 2025. Personal communication via Leek thread.

Freqtrade contributors. Freqtrade: Open-source crypto trading bot, 2026. `https://www.freqtrade.io/en/stable/`.

M. Inan. Predictability of funding rates. *SSRN 5576424*, 2025.

T. Le et al. Funding-aware optimal market making for perpetual DEXs. *arXiv preprint arXiv:2605.06405*, 2026.

Marcos López de Prado and David H. Bailey. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

J. Pang et al. Markov and HMM for regime detection in cryptocurrency markets: Evidence from Bitcoin. *Preprints.org 202603.0831*, 2026.

F. Ravagnani et al. Continuous position shrinkage for regime-sensitive strategies. *arXiv preprint*, 2026. Provides $h_{\text{robust}} = h_{\text{standard}} \cdot (1 - \delta/\sigma^2)$ formulation.

A. Simhi et al. Old habits die hard: Why LLM-generated strategies recycle discredited indicators. *arXiv preprint*, 2026.

A Per-strategy result tables (full leaderboard)

Table 3 (in §5) is the full common-window leaderboard. Per-strategy result cards live in the companion repository under `backtesting/wiki/results/2026-05-*.md`. Each card contains: trade log summary, per-layer metric table, MDB triplet, correlation row, and the pre-registered kill-criteria document referenced.

B MDB definition and computation

Given strategies $\{s_1, \ldots, s_k\}$ with daily log-return series $\{r_t^{(i)}\}_{t=1}^{T}$, define the weighted book return as $r_t^{B,w} = \sum_i w_i \, r_t^{(i)}$ and its annualised Sharpe as

$$\mathrm{SR}^{B,w} = \sqrt{365} \cdot \frac{\bar{r}^{B,w}}{\sigma^{B,w}}. \tag{9}$$

Equal weights: $w_i = 1/k$ trivially.

Risk-parity weights are computed on a trailing 90-day return volatility $\sigma_t^{(i)}$ at each t, with $w_t^{(i)} \propto 1/\sigma_t^{(i)}$ and renormalised so $\sum_i w_t^{(i)} = 1$. The reported MDB-rp uses the endpoint weights for clarity; we have verified that re-computing under rolling weights changes the headline number by less than 0.02.

Mean-variance weights solve

$$\max_w \; w^\top \mu \; - \; \tfrac{1}{2}\lambda \, w^\top \Sigma w \tag{10}$$

$$\text{s.t. } w_i \geq 0, \; \sum_i w_i = 1, \tag{11}$$

with μ, Σ the sample mean and covariance of daily returns and $\lambda = 1$. We acknowledge this is unstable at our N and report MDB-mv as an upper bound only; the robust flag (positive under all three schemes) is the binding gate.

The common window is the intersection of all strategy backtest date ranges ($\geq$ 90 days required). Missing days within a strategy's nominal range are treated as flat ($r = 0$); we have checked that switching to NaN-and-pairwise-complete shifts pairwise correlations by less than 0.05 on average.

C Kill-criteria pre-registration template

Every new strategy in the universe receives a dated decision document under `backtesting/wiki/decisions/00N-k` *before* its first backtest. The template (compressed):

1. Strategy name, family code, universe, timeframe.
2. Hypothesis (one sentence).
3. Reference backtest baseline (if reusing an existing measurement) or `N/A` if first-of-family.
4. **Hard kills (K1–K4):** hard-MDD ceiling, consecutive full-stop trades, rolling-365-day return floor, walk-forward quarterly Calmar floor. Each specified as a scalar with rationale tied to either the reference baseline or a family-level prior.
5. **Continuous shrinkage formula** (Davies–Ravagnani style): size multiplier as a product of factor terms, each clipped to $[0, 1]$.
6. **Re-evaluation triggers:** regime shifts, fee-schedule changes, funding-sign flips, cross-family comparisons. Trigger an explicit written review within 5 trading days; outcome must be one of {continue at current size, force-shrink, hard-kill}.
7. Source-of-truth note: this document wins over any conflicting leaderboard cell until superseded by a new dated decision document.

Decision 004 (Algo-traders project, 2026a) is the canonical worked example. Documents 006, 007, 008 follow the same template for family-specific K1 thresholds (pairs $K1 = 8\%$, cross-sectional $K1_{\text{xs}} = 12\%$, funding-MR $K1_{\text{fmr}} = 5\%$).